Semantic-Assisted Feature Integration Network for Multilabel Remote Sensing Scene Classification

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***Abstract*— With remote sensing (RS) images' resolution increasing, a single scene label cannot adequately represent RS scenes' contents. Therefore, multilabel RS scene classification (MLRSSC) is gradually attracting the researchers' attention. Many methods have been proposed recently, and most use deep features or semantic connections to complete MLRSSC. However, they ignore the combination of these two aspects. In addition, the high interclass similarity and low intraclass similarity of RS images limit the robustness of these methods. In this article, we propose a semantic-assisted feature integration network (SFIN) to overcome the above limitations. It contains a dual-scale feature extractor module (DFEM), a local semantic enhance module (LSEM), a cross-scale interactive attention module (CIAM), and a classifier module (CM). DFEM utilizes the convolutional neural networks (CNNs) to extract multiscale features from RS images. LSEM extracts semantic information and establishes their relationships at different scales. CIAM enhances the feature representation by interacting with the clues across different scales. CM completes the prediction of classification (CLA) results. Integrating them into an end-to-end framework, SFIN can discover the diverse and complex land covers hidden in RS images. Furthermore, to ensure the accuracy of explored semantics and enhance the SFIN's feature extraction ability, we design a semantic supervision (SS) loss and a semantic-based contrastive learning (SB-CL) loss. They are in charge of the correctness and discrimination of the mined semantics. Along with the typical CLA loss, SFIN can be adequately trained. Extensive experiments have been conducted on four MLRSSC datasets, and the positive results demonstrate that SFIN outperforms many existing methods in MLRSSC tasks. Our source codes are available at: https://github.com/TangXu-Group/multilabelRSSC/tree/main/SFIN.**

***Index Terms*— Multilabel, remote sensing (RS) scene classification (CLA).**

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I. Introduction

REMOTE sensing (RS) scene classification (RSSC) has always attracted the attention of researchers due to its widespread applications [1], [2], [3]. With the rapid development of deep learning, many successful methods have been proposed, and they promoted the performance of RSSC to a new level [4], [5], [6]. In the common RSSC tasks, an RS image always has a single scene label. However, this setting is inadequate in the current high-resolution era [7], [8]. As shown in Fig. 1, two RS images are labeled as "Park" and "Port." Nevertheless, there are many other land covers in each image, such as "Dock" and "Buildings." Furthermore, many land covers appear simultaneously in both scenes. In this case, the single scene label is obviously insufficient to represent the RS scenes' contents [9]. Therefore, leveraging multiple semantic labels representing land covers to comprehensively express the complex contents within RS scenes is imperative, so the multilabel (ML) RSSC task has gained more attention from researchers.

Multilabel RSSC (MLRSSC) aims to identify whether the land covers exist in RS scenes or not [10], and numerous methods have been introduced in recent years [11], [12], [13], [14], [15]. As a branch of RSSC, visual features play an important role in MLRSSC. Many works pay attention to extracting or learning discriminative feature representations from RS scenes. We name them feature-based methods for convenience. At the beginning, spatial, texture, and other low-level representations [16], [17], [18] were extracted manually. Along with some traditional classifiers (e.g., support vector machine), acceptable MLRSSC results can be achieved. Currently, convolutional neural networks (CNNs) have dominated the MLRSSC community due to their powerful learning capability and end-to-end learning manner. A large number of models have been developed based on CNN and its varieties [19], and they produce cracking performance. Moreover, with the support of advanced techniques, such as the attention mechanism [20], multiscale information exploration [21], and metric learning [22], the performance of those models has been further enhanced.

Although feature-based methods are feasible, they ignore connections between different semantic objects, which is crucial for MLRSSC [23], [24]. For instance, they hardly explain why the "Dock" and "Water" frequently appear together

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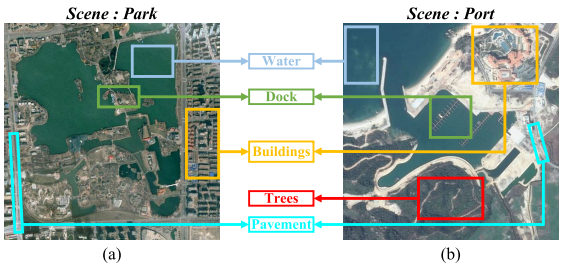


Fig. 1. Here are two RS images. In (a), the scene label is "Park," while in (b), it is "Port." We notice that each image encompasses a diverse array of land covers. In addition, RS images depicting different scenes may share identical land cover types.

but not the "Sand." To address this issue, scholars develop some specific approaches [25], [26], [27], and we name them semantic-based methods. They always incorporate two steps: semantic information extraction and semantic relationship modeling. The former is usually completed using prior knowledge or feature learning tools, and the latter is typically accomplished with the help of sequence relationship models [28], [29]. However, the challenges of low intraclass similarity and high interclass similarity in RS scenes [30] prevent semantic-based methods from reaching the pleasurable results we expected.

To overcome this obstacle, researchers have introduced contrastive learning [31], [32] into MLRSSC tasks. Generally, the prerequisite of contrastive learning is the selection of positive and negative sample pairs. For normal RSSC tasks, the scenes with the same semantic labels are generally treated as positive pairs, while those with different labels are regarded as negative pairs [33]. For MLRSSC, since each RS scene has multiple semantic labels, the scenes with/without overlapping labels are deemed as the positive/negative sample pairs [34], [35], [36], [37]. By pulling in and pushing away the positive and negative sample pairs, RS scenes with low intraclass similarity and high interclass similarity can be distinguished appropriately [38], [39], [40].

Although the above methods work well in MLRSSC, there is still room for improvement. First, the existing methods focus either on feature learning or semantic relationship establishment. However, they ignore the effective integration of these two aspects, which is the key to achieving high-precision MLRSSC. Second, accurately mining the semantics hidden in an RS scene is a prerequisite for establishing semantic relationships. However, due to limited supervision and the randomness of deep learning, the correctness of the semantics explored from RS scenes is questionable, leading to a decrease in the credibility of the established semantic relationships. Therefore, the accuracy of MLRSSC would be whittled down. Third, for some MLRSSC models, the conventional contrastive learning is introduced and conducted at the image level. They focus on the overlapped semantics in various RS scenes but disregard the unique semantics in different RS scenes. Those exclusive semantics make the final results cannot meet what we expected. For example, in Fig. 1, two scenes should be pulled together as they share four common semantics. However, the "Trees" will be sacrificed in this way, negatively impacting the classification (CLA) results.

Based on the above discussion, we propose a new model named semantic-assisted feature integration network (SFIN) for MLRSSC in this artoicle. SFIN involves a dual-scale feature extractor module (DFEM), a local semantic enhance module (LSEM), a cross-scale interactive attention module (CIAM), and a classifier module (CM). Their functions are to extract multiscale features, discover semantics and build their connections, facilitate information exchange between various scales and complete CLA tasks, respectively. To push SFIN to pay attention to the diverse semantics hidden in RS scenes and accomplish CLA properly, three loss functions are designed: a semantic supervision (SS) loss, a semantic-based contrastive learning (SB-CL) loss, and a CLA loss. The SS loss ensures the category-specificity and correctness of the extracted semantics. This way, the connections between different land covers can be established accurately. The SB-CL loss enhances the discrimination of the obtained semantics. As a result, the diverse land covers can be distinguished easily. By applying the above two losses to LSEM, the capability of SFIN to identify complex contents of RS scenes can be improved. Along with the typical CLA loss, SFIN can be trained for MLRSSC tasks.

The main contributions of this article are as follows.

1) An end-to-end network named SFIN is proposed. By mining the multiscale clues, exploring the diverse land covers, analyzing their relationships, and enhancing the feature representation, our SFIN can fully understand the complex contents within RS images for the ML CLA.
2) Under deep supervision and contrastive learning paradigms, the SS and SB-CL losses are developed and applied to LSEM to help SFIN acquire accurate and discriminative semantic information from RS images. Combining them with a typical CLA loss, our SFIN can be appropriately trained for accurate ML CLA results.
3) Comprehensive experiments have been conducted on the four popular datasets. The positive numerical and visual results confirm that the proposed SFIN is helpful in MLRSSC tasks. Compared with many existing methods, SFIN can achieve better results.

II. Related Works

A. Feature-Based MLRSSC Methods

Feature-based methods focus on obtaining and fully using the discriminative visual features for the final CLA. Due to the strong feature learning capacity and specific training manner, CNN has dominated this field in recent years [41]. As early as 2019, Stivaktakis et al. [42] constructed a simple CNN to complete MLRSSC tasks. Along with the valuable data augmentation scheme, positive results can be produced in the case of insufficient samples. With the development of CNN, a growing number of advanced CNNs have been proposed, and their performance in MLRSSC is also impressive [19]. For example, Zhu et al. [43] proposed an MLRSSC framework consisting of a CLA branch and an embedding branch. The CLA branch is composed of Inception-ResNet-v2 [44], which integrates spatial attention mechanism and skip connection.

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The embedding branch adopts a triplet loss to ensure similar RS scenes have similar feature representations. By combining two branches, the representative features can be generated under the influence of multiscale information and salient region information. However, CNN's shortcomings in capturing global context information limit its development. To this end, a global context-based multilevel feature fusion network (GCMLFNet) was proposed in [45]. In addition to the basic CNN and a multiscale feature fusion module, GCMLFNet introduces a global context module to establish long-distance dependencies between different pixels. Similarly, graph convolutional network (GCN) is used in [46] to enhance CNN's feature learning from the global aspect for MLRSSC. Moreover, transformer [47] and its variants [48] are also frequently employed to enhance the model's capability for processing global information. While effective, these approaches typically come with significant computational costs. Therefore, some lightweight sequence relationship modeling models (such as Mamba [49]) show great potential in RS scenarios. In addition, unlike the above methods, which pay attention to changing the network structure, Kang et al. [50] proposed a generic loss function for MLRSSC named scalable neighbor discriminative loss (SNDL). SNDL aims to adjust the distances of RS images according to the degree of overlap between labels to generate better feature representations. Experimental results confirm that SNDL is flexible for different CNNs without additional cost.

B. Semantic-Based MLRSSC Methods

The core idea of semantic-based methods is to capture the semantics from RS scenes and analyze their correlations. Initially, scholars were accustomed to manually defining and extracting semantic relationships from the dataset for subsequent learning and CLA. For example, Huang et al. [51] proposed a label co-occurrence matrix to complete MLRSSC tasks. The matrix is constructed based on the labels of the original dataset. The semantic links among various labels can also be represented by this matrix. Then, the visual features learned by CNNs are used to classify the results with the help of semantic connections. Another similar approach was proposed in [27]. Based on the label co-occurrence matrix, GCN is used to fine-tune the label co-occurrence matrix to explore advanced connections further. Although feasible, manually extracting semantic information is resource-intensive with low transferability. Therefore, human and big data knowledge are used to enhance the accuracy of manual semantic information in [52]. In addition, many methods have been proposed to explore the semantics and build their relations automatically and simultaneously [25], [53], [54]. For instance, a relation network was introduced in [25] for MLRSSC. It first alters the final layer of visual geometry group (VGG) [55] to obtain the features associated with different semantics. Then, a learnable affine transformation matrix is utilized to identify crucial spatial semantic areas and eliminate redundant information. Finally, multiple fully connected layers are connected in parallel to explore semantic relationships and complete CLA tasks. To further enhance the effectiveness of semantic relationships, advanced sequence relationship models have been introduced into MLRSSC, with transformer being a notable example used in the semantic relation inference network (SR-Net) [26]. After discovering the semantics within RS scenes, SR-Net uses several transformer encoders to establish connections between different semantics and designs a semantic attention mechanism to enhance the obtained semantic information. Positive results confirm the usefulness of SR-Net.

C. Multilabel Contrastive Learning

Contrastive learning facilitates the reduction of distances among positive samples within the feature space while increasing the gaps between negative samples. It enables the model to acquire superior feature representations from images by analyzing their distributions [56]. Extensive experiments in [34] and [35] reveals that treating samples with overlapping labels as positive pairs leads to the learning of satisfied visual feature representations. Therefore, some researchers extend contrastive learning to the domain of ML learning. For instance, Qian et al. [36] developed a similarity approach to select the positive sample for ML contrastive learning. Considering the class imbalance issue, the authors also design a label similarity measure to make the model focus on sample pairs with lower semantic consistency. Transferring the above concept to ML CLA, an ML contrastive learning framework was introduced in [37]. An ML contrastive learning loss function and a novel hierarchy-preserving loss are designed to optimize the feature space and maintain the hierarchical relationships between categories. This approach not only overcomes the limitations associated with contrastive learning when relying on single-label but also affirms the effectiveness of ML contrastive learning. Another similar work was proposed in [57] by combining supervised and unsupervised contrastive learning. The positive and negative samples generated by different data augmentations improve the model's understanding of RS scenes. Although feasible, the aforementioned methods remain limited to the image-level understanding of RS scenes, constraining the model's ability to identify unique land cover types.

III. Proposed Method

A. Overall Framework

Before illustrating our method, the preliminaries of MLRSSC are introduced first. Assuming that $\mathbf{I}$ represents one of the images in a mini-batch ML RS set D with capacity N, its corresponding label $\mathbf{y} \in \{0, 1\}^c$ can be defined as follows: if $\mathbf{I}$ contains the land cover belonging to category j, $\mathbf{y}^j$ is 1, otherwise, it is 0, where c means the number of semantic categories.

The schematic framework of our SFIN is exhibited in Fig. 2, which consists of a DFEM, an LSEM, a CIAM, and a CM. DFEM intends to extract multiscale features from RS images. LSEM explicitly establishes relationships between RS image pixel areas and various semantics (depicted by the orange solid lines in Fig. 2) and implicitly builds connections among different semantics (illustrated by the orange dotted lines in Fig. 2). CIAM enhances and supplements

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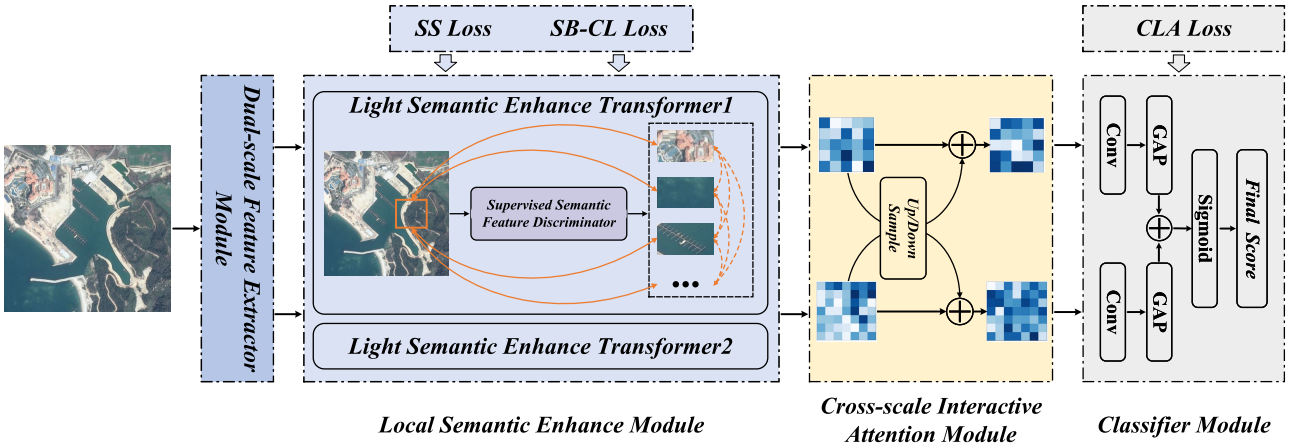


Fig. 2. Framework of SFIN, which consists of a DFEM, an LSEM, a CIAM, and a CM. In addition, an SS loss, an SB-CL loss, and a CLA loss are used to optimize SFIN. They are responsible for ensuring the accuracy of semantic information, enhancing the discriminability between different semantics, and completing the CLA task.

information at different scales through interactive attention maps. CM integrates CLA results from various scales and generates the final prediction. The proposed SFIN is jointly trained using three loss functions: SS loss, SB-CL loss, and CLA loss. The SS loss and SB-CL loss ensure the reliability and discrimination of the semantic information extracted by LSEM. The CLA loss guides CM in accomplishing MLRSSC tasks. Now, we introduce the structure of SFIN for the input $\mathbf{I}$ and the loss functions on the mini-batch set D in detail.

B. Structure of the Proposed SFIN

1) Dual-Scale Feature Extractor Module: RS images often encompass land covers with the same semantics but diverse scales, making it challenging for single-scale features to represent complex contents comprehensively. To this end, DFEM is proposed. Considering that the multiscale features learned by the hierarchical CNNs contain rich information [58], i.e., the features at shallow scale focus on low-level messages such as texture and color, while the features at deep scale involve high-level contextual clues, DFEM selects ResNet18 [59] containing four residual layers as the backbone to capture both low- and high-level features from $\mathbf{I}$. Since the feature maps output by the first two layers tend to contain redundant information [58], we exclusively use the outputs from the last two layers. For convenience, we mark them as $\mathbf{F}_1 \in \mathbb{R}^{H_1 \times W_1 \times C_1}$ for the last one and $\mathbf{F}_2 \in \mathbb{R}^{H_2 \times W_2 \times C_2}$ for the penultimate one.

2) Local Semantic Enhance Module: LSEM is introduced to explore the semantics and construct their connections to enhance the model's ability to identify different land covers at the semantic aspect. As exhibited in Fig. 2, LSEM contains two parallel light semantic enhance Transformers (LSETs) to handle $\mathbf{F}_1$ and $\mathbf{F}_2$, respectively. They have the same structure but do not weight-sharing. For clarity, we select one of them and unify the input data as $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$ to explain LSET in detail. The architecture of LSET is shown in Fig. 3(a), which incorporates a supervised semantic feature discriminator (SSFD), an attention-based feature updating mechanism (AFUM), and a multilayer convolution block (MCB). SSFD is developed to accurately extract category-aware semantics from the local feature map $\mathbf{X}$. AFUM is designed to build the relations between feature pixels and semantics immediately. Meanwhile, the connections among various semantics can be indirectly established using feature pixels as a medium. MCB is a simple yet powerful submodule aiming to exploit local information further.

Fig. 3(b) displays the SSFD structure, which has two parallel branches. In the first branch, a 1×1 convolutional layer with a sigmoid function is first applied on $\mathbf{X}$ to generate class attention map $\mathbf{CAM} \in \mathbb{R}^{H \times W \times c}$. Following [26], [54], and [60], the ith channel of $\mathbf{CAM}$ represents the contribution score of $\mathbf{X}$ to the ith category. Then, the elementwise multiplication is conducted on $\mathbf{X}$ and each channel of $\mathbf{CAM}$, followed by the global average pooling (GAP) to construct the unique semantic information vector for each class. The above process can be formulated as follows:

$$\begin{aligned} \mathbf{CAM} &= \sigma(\mathrm{conv}_{1\times1}(\mathbf{X})) = [\mathbf{CS}^1, \ldots, \mathbf{CS}^c] \\ \mathbf{SF} &= [\mathbf{sf}^1, \ldots, \mathbf{sf}^c]^{\mathrm{T}}, \quad \mathbf{sf}^j = \mathrm{GAP}(\mathbf{X} \otimes \mathbf{CS}^j) \end{aligned} \tag{1}$$

where $\mathrm{conv}_{1\times1}(\cdot)$, $\sigma(\cdot)$, $\otimes$, and $\mathrm{GAP}(\cdot)$ denotes the 1×1 convolution, sigmoid function, elementwise multiplication, and the GAP function, respectively, $\mathbf{CS}^j \in \mathbb{R}^{H \times W}$ indicates the jth class contribution score, $\mathbf{sf}^j \in \mathbb{R}^C$ denotes the semantic information vector for the jth class, and $\mathbf{SF} \in \mathbb{R}^{c \times C}$ represents the extracted semantics. In the second branch, we orderly pass $\mathbf{X}$ through a GAP and the 1×1 convolutional layer to obtain class activation vector $\mathbf{r} \in \mathbb{R}^c$. It will be used to guarantee the accuracy of $\mathbf{CAM}$. The details are shown in Section III-C1.

After obtaining the semantics, the next step is to model their connections. The common methods [25], [26], [50] are accustomed to applying sequence relationship methods [28], [61] on semantic information vectors to achieve the relationships. Also feasible, they ignore the information hidden in the feature pixels, which limits the contributions of semantic relationships to the scene understanding. To overcome this shortcoming, we develop AFUM to establish semantic relationships with the assistance of feature pixels. This way, the messages within the global semantics and local feature pixels can be considered

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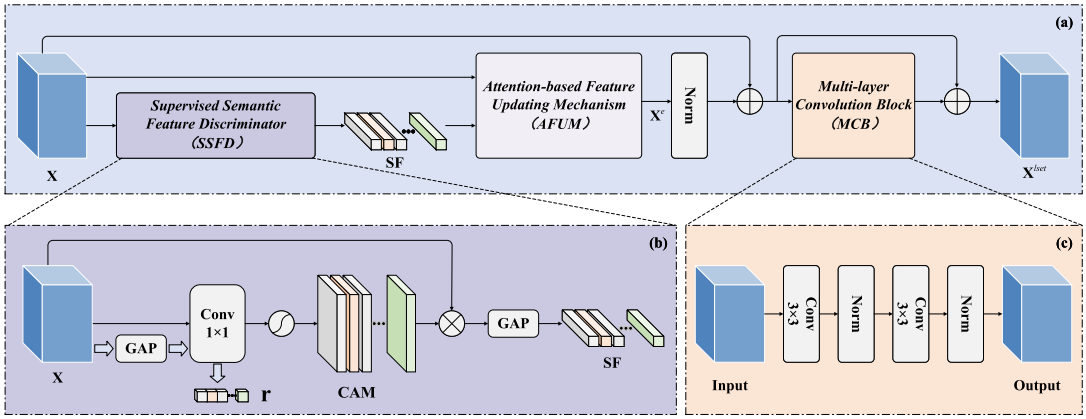


Fig. 3. (a) Overall framework of LSET, which mainly consists of an SSFD, an AFUM, and an MCB. SSFD aims to extract category-aware semantics, and its structure is shown in (b). AFUM is designed to build the relations between pixel-level features and semantics. MCB is responsible for further exploiting local information, and its structure is shown in (c).

simultaneously during the connection building, ensuring the generated semantic associations are good and credible for interpreting RS scenes.

Particularly, with the inspiration of vision transformer (ViT) [47], we apply the self-attention mechanism to the original feature map $\mathbf{X}$ and obtained $\mathbf{SF}$ to explore the relations between feature pixels and semantics directly. This step can be formulated as follows:

$$\mathbf{X}'_{i,j} = \mathbf{M}_{i,j} \begin{bmatrix} \mathbf{sf}^1 \\ \dots \\ \mathbf{sf}^c \end{bmatrix} \mathbf{W}^V$$
$$\mathbf{M}_{i,j} = \phi\left(\mathbf{X}_{i,j}\mathbf{W}^Q \left[\mathbf{sf}^1, \dots, \mathbf{sf}^c\right] \mathbf{W}^K / \sqrt{d}\right) \quad (2)$$

where $\phi(\cdot)$ represents the softmax function, $\mathbf{W}^Q$, $\mathbf{W}^Q$, and $\mathbf{W}^Q$ denote the three learnable projection matrixes, $\mathbf{X}_{i,j}$ and $\mathbf{X}'_{i,j}$ denote the ith row and jth column element of the original feature map and the enhanced feature map, $\sqrt{d}$ is a scaled factor, and $\mathbf{M}_{i,j}$ is a weight matrix that contains the correlations between $\mathbf{X}_{i,j}$ and various semantics. Here, similar to multihead self-attention of ViT, we execute h AFUMs and combine their outputs to acquire the ultimate enhanced results $\mathbf{X}^e \in \mathbb{R}^{H\times W\times C}$. When we get $\mathbf{X}^e$, since the relations between feature pixels and semantics have been decided, the connections between various semantics can also be obtained indirectly.

Next, we pass the normalized $\mathbf{X}^e$ through MCB to further mine local information. MCB is composed of two consecutive pairs of 3×3 convolutional and batch norm layers [as shown in Fig. 3(c)]. In addition, the residual structure is employed to mitigate information loss. After MCB, the output of LSET $\mathbf{X}^{\text{lset}} \in \mathbb{R}^{H\times W\times C}$ can be obtained. Based on the above discussion, after passing $\mathbf{F}_1$ and $\mathbf{F}_2$ through two LSETs, they can be mapped into $\mathbf{F}_1^{\text{lset}} \in \mathbb{R}^{H_1\times W_1\times C_1}$ and $\mathbf{F}_2^{\text{lset}} \in \mathbb{R}^{H_2\times W_2\times C_2}$, which are rich in local, pixel-sematic, and semantic-semantic clues.

3) Cross-Scale Interactive Attention Module: Considering the features from different scales enclose diverse clues, CIAM is introduced to enhance them by mutually interacting with the valuable knowledge across scales. This way, the contents of RS images can be fully explored at different scales. The schematic of CIAM is shown in Fig. 4. First, a max pooling and an average pooling (AP) are simultaneously applied to $\mathbf{F}_1^{\text{lset}}$ and $\mathbf{F}_2^{\text{lset}}$ to reduce the channel dimension. Meanwhile, two 3×3 convolutions, along with the sigmoid function, are conducted to produce the attention scores $\mathbf{A}_1 \in \mathbb{R}^{H_1\times W_1}$ and $\mathbf{A}_2 \in \mathbb{R}^{H_2\times W_2}$ for feature map pixels across various scales in the spatial dimension. This process can be expressed as follows:

$$\mathbf{F}_j^{\text{lset}'} = \text{concat}\left(\text{MP}\left(\mathbf{F}_j^{\text{lset}}\right), \text{AP}\left(\mathbf{F}_j^{\text{lset}}\right)\right), \quad j = 1, 2$$
$$\mathbf{A}_j = \sigma\left(\text{conv}_{3\times 3}\left(\mathbf{F}_j^{\text{lset}'}\right)\right), \quad j = 1, 2 \quad (3)$$

where $\text{MP}(\cdot)$, $\text{AP}(\cdot)$, $\text{conv}_{3\times 3}(\cdot)$, and $\text{concat}(\cdot)$ denote the maximum pooling, AP, 3×3 convolution, and concatenation functions, respectively, and $\mathbf{A}_j$ refers to the attention scores for $\mathbf{F}_j^{\text{lset}}$. Then, we upsample $\mathbf{A}_1$ and add it with $\mathbf{A}_2$ to obtain a novel attention map, which is then element-wise multiplied with $\mathbf{F}_2^{\text{lset}}$ to get the ultimate feature representation. Similarly, the downsampled $\mathbf{A}_2$ is added to $\mathbf{A}_1$ and element-wise multiplied with $\mathbf{F}_1^{\text{lset}}$. This process can be formulated as follows:

$$\mathbf{F}_1^{\text{ciam}} = (\text{AP}(\mathbf{A}_2) + \mathbf{A}_1) \otimes \mathbf{F}_1^{\text{lset}}$$
$$\mathbf{F}_2^{\text{ciam}} = (\text{NNI}(\mathbf{A}_1) + \mathbf{A}_2) \otimes \mathbf{F}_2^{\text{lset}} \quad (4)$$

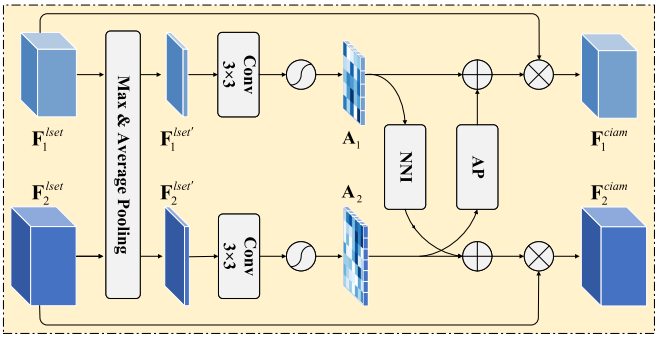


Fig. 4. Schematic of CIAM. In particular, features at different scales ($\mathbf{F}_1^{\text{lset}}$, $\mathbf{F}_2^{\text{lset}}$) undergo max and AP, convolution, and sigmoid activation functions to generate attention scores ($\mathbf{A}_1$, $\mathbf{A}_2$). Next, $\mathbf{A}_1$ is upsampled using NNI and added to $\mathbf{A}_2$, while $\mathbf{A}_2$ is downsampled using AP and added to $\mathbf{A}_1$. The new attention scores are multiplied with the original features, facilitating cross-scale information interaction and enhancement.

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where $\mathbf{F}_1^{\text{ciam}} \in \mathbb{R}^{H_1 \times W_1 \times C_1}$ and $\mathbf{F}_2^{\text{ciam}} \in \mathbb{R}^{H_1 \times W_1 \times C_1}$ denote the features for CLA, and AP$(\cdot)$ and NNI$(\cdot)$ denote the AP for the downsampling operation and nearest-neighbor interpolation (NNI) for upsampling operation, respectively.

Effective information interaction between two scale features is achieved through CIAM. This interactive design facilitates collaboration between scales to refocus on areas overlooked during the forward propagation process. At the same time, this mechanism deepens the understanding of the features' content and helps alleviate the impact of redundant information.

4) Classifier Module: At the top of SFIN, CM consisting of two classifiers is used to generate CLA results based on $\mathbf{F}_1^{\text{ciam}}$ and $\mathbf{F}_2^{\text{ciam}}$. Each classifier includes a 1×1 convolutional layer and a GAP layer, and the number of output channels of the convolutional layer equals the number of categories c. Finally, the results of the two classifiers are added and activated through a sigmoid function to obtain the final score $\mathbf{FS} \in \mathbb{R}^c$ for each class.

C. Training Losses for the Proposed Network

1) SS Loss: As discussed in Section III-B2, the correctness of the semantics is crucial for ML scene understanding, which is directly related to the generated class attention map **CAM**. Therefore, we first develop the SS loss to ensure SSFD in LSET can achieve the proper semantics corresponding to class activation vectors. Suppose the RS scenes in a mini-batch have been mapped into $\{\mathbf{r}_1^1, \ldots, \mathbf{r}_N^1\}$ and $\{\mathbf{r}_1^2, \ldots, \mathbf{r}_N^2\}$ by the two LSETs. Then, the accuracy of class attention maps **CAM**s can be improved by reducing the gaps between class activation vectors and labels $\{\mathbf{y}_1, \ldots, \mathbf{y}_N\}$. To this end, we adopt the binary-cross-entropy (BCE) loss function [62], which can be formulated as follows:

$$\begin{aligned} \mathcal{L}_{\text{ss}} &= \sum_{i=1}^{N} \text{BCE}\left(\mathbf{r}_i^1, \mathbf{y}_i\right) + \text{BCE}\left(\mathbf{r}_i^2, \mathbf{y}_i\right) \\ \text{BCE}(\mathbf{r}, \mathbf{y}) &= -\sum_{j=1}^{c} \mathbf{y}_j \log\left(\sigma(\mathbf{r}_j)\right) + \left(1 - \mathbf{y}_j\right) \log\left(1 - \sigma(\mathbf{r}_j)\right) \end{aligned} \tag{5}$$

where $\mathcal{L}_{\text{ss}}$ pushes each channel of the feature map toward a specific category [63], [64], resulting in class activation maps that can accurately highlight the areas of interest for each category. This way, the certainty of the extracted semantics can be ensured, laying a solid foundation for establishing definite semantic connections.

2) SB-CL Loss: To overcome the challenge of low intra-class similarity and high interclass similarity among land covers in RS scenes, and to enhance the model's ability to distinguish them, we propose the SB-CL loss under the framework of contrastive learning. Traditional contrastive learning typically adjusts sample distances at the image level. However, multiple semantics within the same image restrict their development in MLRSSC. Hence, we extend contrastive learning to the semantic level, benefitting from the extracted class-special semantic information vectors from SSFD. Overall, we treat information vectors with the same semantics as positive pairs and those with different semantics as negative pairs. By narrowing/enlarging the gaps between positive/negative pairs, the discrimination of the learned semantics can be improved. For better understanding, we take the "Buildings" in Fig. 1 as positive samples and other objects as negative samples and illustrate the target of SB-CL loss in Fig. 5.

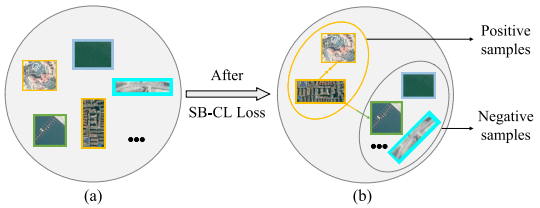


Fig. 5. Schematic interpretation of the SB-CL loss function. It aims to cluster/separate the land covers with the same/different semantics in the feature space. (a) Distribution of different land covers in the original feature space. (b) Distribution of different land covers in the feature space after SB-CL loss function.

For a mini-batch RS scenes D, assume that their semantics $\{\mathbf{SF}_1^1, \ldots, \mathbf{SF}_N^1\}$ and $\{\mathbf{SF}_1^2, \ldots, \mathbf{SF}_N^2\}$ have been extracted by SSFD of the two LSETs. Taking the rich scale clues into account, two sets of semantics are concatenated to generate dual-scale semantic vectors $\{\mathbf{DSF}_1, \ldots, \mathbf{DSF}_N\}$, where $\mathbf{DSF}_i = [\mathbf{dsf}_i^1, \ldots, \mathbf{dsf}_i^c]^{\text{T}} \in \mathbb{R}^{c \times (C_1 + C_2)}$. Then, we construct a single-label semantic dataset SD based on the labels of RS scenes in this batch

$$\text{SD} = \{\text{SD}_1, \ldots, \text{SD}_c\}, \quad \text{SD}_j = \left\{ l_{\mathbf{y}_i^j = 1} \cdot \mathbf{dsf}_i^j \right\}_{i=1}^{N} \tag{6}$$

where SD_j denotes the subdataset for category j (i.e., the semantics within this subdataset are labeled as j), and $l_\zeta \in \{0, 1\}$ is an indicator function, if the condition ζ is met, $l_\zeta = 1$, otherwise, $l_\zeta = 0$. Considering the scenario where some classes are absent in an RS image, the extracted semantics may tend to be the background. To mitigate this issue, we only select the semantics within the RS images based on the labels. Then, the SB-CL loss is formulated to minimize/maximize the distances between semantics belonging to the same/different subdatasets. Its definition is

$$\begin{aligned} \mathcal{L}_{\text{sbcl}} &= -\sum_{a \in \text{SD}} \frac{1}{|P(a)|} \mathcal{L}_{\text{sbcl}}^a \\ \mathcal{L}_{\text{sbcl}}^a &= \sum_{p \in P(a)} \log\left(\frac{\exp\left(\frac{d(a,p)}{\beta}\right)}{\exp\left(\frac{d(a,p)}{\beta}\right) + \sum_{n \in N(a)} \exp\left(\frac{d(a,n)}{\beta}\right)} \right) \end{aligned} \tag{7}$$

where a denotes the anchor sample with semantic label j in SD, $\{P(a) = \text{SD}_j \backslash \{a\}\}$ and $\{N(a) = \text{SD} \backslash \text{SD}_j\}$ refer to the positive and negative sample sets of a, $p \in P(a)$ and $n \in N(a)$ indicate related positive and negative samples, $|P(a)|$ means the volume of positive set, β is the temperature coefficient, and $d(\cdot, \cdot)$ denotes the cosine similarity. The semantic-level SB-CL loss function aids the model in capturing highly discriminative semantic information, facilitating a better comprehension of the distinctions and correlations among various semantic categories. This enables our model to have a deeper understanding of the characteristics and distribution of different land covers in RS images, thereby improving the model's accuracy in identifying different land covers.

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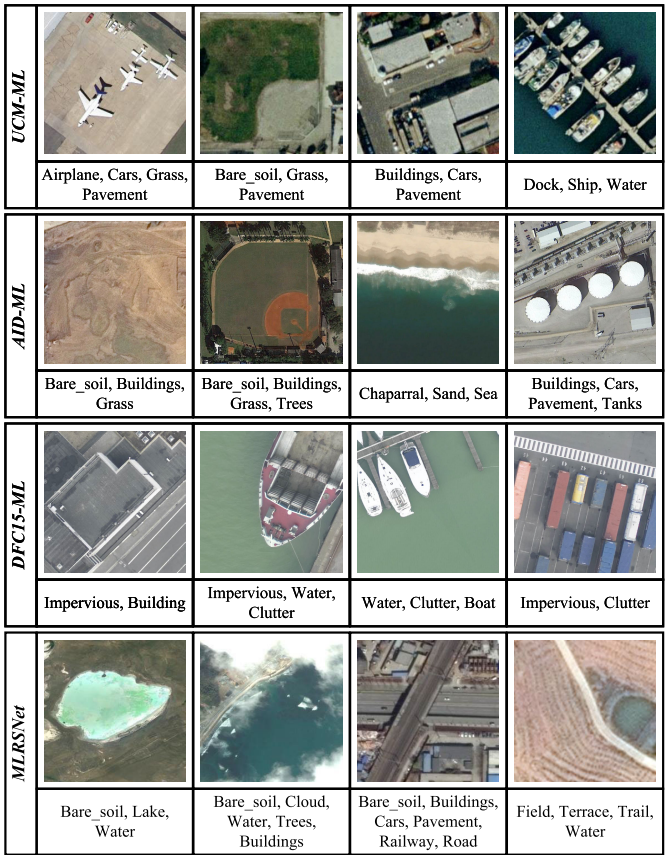


Fig. 6. Examples of different datasets. In each block, the RS images are shown at the top, and their multiple semantic labels are at the bottom.

3) CLA Loss: In the end, we use the BCE loss function as our CLA loss to reduce the gap between the final score **FS** (generated by CM) and the label **y**. The formula is as follows:

$$\mathcal{L}_{\text{cla}} = \sum_{i=1}^{N} \text{BCE}(\mathbf{FS}_i, \mathbf{y}_i). \tag{8}$$

Based on the above discussion, the overall loss function for SFIN can be defined as follows:

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{cla}} + \mathcal{L}_{\text{ss}} + \lambda \mathcal{L}_{\text{sbcl}} \tag{9}$$

where λ is a hyperparameter. Since $\mathcal{L}_{\text{cla}}$ and $\mathcal{L}_{\text{ss}}$ are developed for CLA and $\mathcal{L}_{\text{sbcl}}$ is designed for contrastive learning, we only add λ on $\mathcal{L}_{\text{sbcl}}$ to balance different terms. The three loss functions work together to ensure that SFIN can correctly extract semantic information, improve the model's discriminative ability to different land covers, and complete MLRSSC tasks.

IV. Experiments

A. Datasets Introduction

To validate the effectiveness of SFIN, we choose four popular datasets tailored for MLRSSC, including the UC Merced ML (UCM-ML) [16], aerial image database ML (AID-ML) [53], data fusion contest 2015 ML (DFC15-ML) [54], and a large-scale high-resolution multi-label remote sensing (MLRSNet) [65] datasets. Their examples are shown in Fig. 6.

1) UCM-ML: The RS images in UCM-ML are derived from the original UCM dataset [66], so it contains 2100 images with a size of 256 × 256 and a pixel resolution of 0.3 m. Each RS image is reassigned 1–7 semantic-level labels from an object set comprising 17 classes. These classes include airplanes, bare_soil, buildings, cars, chaparral, court, dock, field, grass, mobile_home, pavement, sand, sea, ship, tanks, trees, and water.

2) AID-ML: AID-ML is constructed based on AID single-label CLA dataset [67], collected from Google Earth imagery of various countries. There are 10 000 high-resolution aerial images in the AID dataset. Their sizes are 600 × 600, and their spatial resolutions vary from 0.5 to 8 m. The RS images in AID are divided into 30 classes, and the volume of images in each category ranges from 220 to 420. AID-ML selects 100 images from each AID category and assigns them semantic-level labels from the same object set as UCM-ML. In addition, the number of labels per image varies from 1 to 11.

3) DFC15-ML: Hua et al. [54] constructed DFC15-ML by aggregating the labels of each pixel into multiple semantic-level labels based on a semantic segmentation dataset. It consists of 3342 RS images acquired by an airborne sensor with a spatial resolution of 5 cm over Zeebrugge, Belgium. The sizes of RS images in DFC15-ML are 600 × 600, each of which has 2–6 labels such as impervious, water, produced in clutter, vegetation, building, tree, boat, and car.

4) MLRSNet: MLRSNet [65] involves 109 161 RS images with sizes of 256 × 256 and diverse resolutions ranging from 10 to 0.1 m. There are 60 predefined semantic labels (such as road, grass, and cars), and the number of semantic categories associated with each RS image varies from 1 to 13.

B. Experimental Details

1) Experimental Settings: All experiments were conducted using the PyTorch framework on an NVIDIA Titan XP GPU with 12 GB of memory. The presented numerical scores are the average results of five experiments. The ResNet18 of SFIN is initialized using the pre-trained weights obtained on the ImageNet dataset, and the other parts of SFIN are initialized randomly. The Adam algorithm is selected to optimize the parameters of SFIN. The training epochs and learning rate are set to be 100 and 4e−5. After the first 50 epochs, the learning rate would be halved. For UCM-ML and AID-ML, 80% of RS images are selected to construct the training set, and the remaining 20% RS images are used as the testing set [25], [65]. For DFC15-ML, we use the dataset partition scheme mentioned in [54], i.e., 2674 for training and 668 for testing. For MLRSNet, according to [65], 20% of the RS images are randomly selected to train SFIN, and the rest of the RS images are used to test. All RS images from different datasets are resized into 256 × 256 before inputting them into SFIN. Also, standard augmentation techniques, such as random horizontal flip, random rotation, and normalization, are applied to stabilize the training process. Three free parameters in our SFIN should be set in advance, including the number of AFUMs h, temperature coefficient β (7), and term balance parameter λ (9). For h, its value is fixed as 4 for different datasets in line with [4]. For the other two parameters,

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TABLE I
PERFORMANCE OF DIFFERENT METHODS COUNTED ON UCM-ML AND AID-ML (%). THE BEST VALUES ARE IN BOLD, AND THE SECOND-BEST VALUES ARE UNDERLINED

Methods	UCM-ML						AID-ML					
	P_e	R_e	$F1_e$	P_c	R_c	$F1_c$	P_e	R_e	$F1_e$	P_c	R_c	$F1_c$
SNDL-BCE [50]	87.24	85.37	86.30	91.27	83.04	86.97	90.06	87.44	88.73	79.81	69.43	74.26
ResRF [11]	86.20	81.49	83.78	92.09	70.48	79.85	89.64	86.82	88.21	79.07	61.83	69.39
GCMLFNet [45]	88.20	81.86	84.91	89.36	82.25	85.66	89.14	88.56	88.85	79.73	69.15	74.06
AL [69]	86.35	86.61	86.48	88.68	80.67	84.49	88.63	88.49	88.56	77.19	70.76	73.85
ForestViT [61]	88.82	83.99	86.34	92.67	74.94	82.87	90.23	88.86	89.53	84.46	68.63	75.72
LD-GCN [27]	86.61	84.34	85.46	90.62	78.09	83.89	90.14	85.75	87.89	79.14	64.77	71.24
BoostLU [70]	89.02	86.86	87.93	92.98	84.21	88.37	89.98	89.22	89.60	82.62	**71.28**	76.53
SR-Net [26]	88.83	86.37	87.59	92.25	84.58	88.25	90.17	88.59	89.37	82.83	69.14	75.37
Ours	**90.33**	**87.30**	**88.79**	**93.05**	**86.46**	**89.63**	**90.26**	**89.45**	**89.85**	**84.74**	70.90	**77.20**

we empirically set $\beta = 0.5$ and $\lambda = 0.1$. Also, their influence on SFIN will be discussed in Section IV-D.

2) Evaluation Metrics: Following [26], [54], and [68], we choose six assessment indicators to assess the proposed SFIN: example-based precision (P_e), example-based recall (R_e), example-based $F1$ score ($F1_e$), class-based precision (P_c), class-based recall (R_c), and class-based $F1$ score ($F1_c$). P_e, R_e, and $F1_e$ evaluate the CLA performance of the model from the perspective of samples, while P_c, R_c, and $F1_c$ from the perspective of categories. The six indicators can be defined as follows:

$$P_e = \frac{1}{N}\sum_{i=1}^{N}\frac{\mathrm{TP}_e^i}{\mathrm{TP}_e^i + \mathrm{FP}_e^i}, \quad R_e = \frac{1}{N}\sum_{i=1}^{N}\frac{\mathrm{TP}_e^i}{\mathrm{TP}_e^i + \mathrm{FN}_e^i}$$
$$P_c = \frac{1}{c}\sum_{i=1}^{c}\frac{\mathrm{TP}_c^i}{\mathrm{TP}_c^i + \mathrm{FP}_c^i}, \quad R_c = \frac{1}{c}\sum_{i=1}^{c}\frac{\mathrm{TP}_c^i}{\mathrm{TP}_c^i + \mathrm{FN}_c^i}$$
$$F1_e = \frac{2 \times P_e \times R_e}{P_e + R_e}, \quad F1_c = \frac{2 \times P_c \times R_c}{P_c + R_c} \tag{10}$$

where TP_e, FP_e, and FN_e denote the true positive, false positive, and false negative values of each sample, respectively, and TP_c, FP_c, and FN_c indicate the true positive, false positive, and false negative values of each class, respectively.

Following [26] and [54], in the inference stage, if the score of a specific category exceeds 0.5, it is determined that the category exists; otherwise, it is determined that it does not exist.

C. Compare With Other Methods

To verify the effectiveness of SFIN, we select the following eight comparison algorithms, they are, SNDL-BCE [50], ResRF [11], GCMLFNet [45], AL [69], ForestViT [61], LD-GCN [27], BoostLU [70], and SR-Net [26]. Among them, AL and BoostLU are general-purpose, and others are RS-oriented. All comparison experiments were conducted by ourselves using the codes released by the authors. Their free parameters were set according to the original literature, and the experimental settings (such as training and testing data partition) are in line with the description in Section IV-B1. Consistent with [4], [25], and [54], we employed ResNet18 as the backbones for all compared methods, except for [61], which is a convolution-free model.

1) Results on UCM-ML and AID-ML: The numerical results of different methods counted on UCM-ML and AID-ML are shown in Table I, with the best results in bold and the second-best ones underlined. After careful observation, it is evident that our SFIN is the most superior model. Taking UCM-ML as an example, our SFIN surpasses other methods in $F1_e/F1_c$ are 2.49%/2.66% (over SNDL-BCE), 5.01%/9.78% (over ResRF), 3.88%/3.97% (over GCMLFNet), 2.31%/5.14% (over AL), 2.45%/6.76% (over ForestViT), 3.33%/5.74% (over LD-GCN), 0.86%/1.25% (over BoostLU), and 1.20%/1.38% (over SR-Net). The enhancements in $F1_e$ indicate that SFIN can precisely identify various land covers within RS images. These encouraging results can be attributed to two factors. First, our model comprehensively understands RS images from various perspectives by leveraging multiscale information. Second, a specialized mechanism for enhancing semantic information ensures that the feature map utilized for CLA encompasses local convolutional information and semantic connections information. The diversity of information lays the foundation for the SFIN's satisfactory performance. In addition, the advancements in $F1_c$ reveals that our method is insensitive to the class imbalance issue inherent in the two datasets. Benefiting from the SS and SB-CL losses, the correctness and discriminability of the semantic information obtained from RS images are ensured, thereby enhancing the overall behavior of SFIN. An interesting observation is that BoostLU stands out as the best model in R_c on AID-ML. This is because of its proficiency in recognizing severely biased categories within the dataset. The class imbalance issue is particularly prominent in AID-ML, especially for "Mobile_home" that only appears once in the training set [54]. Fortunately, our SFIN achieves the second-best R_c values, and the gap between it and BoostLU remains within an acceptable range. This proves the effectiveness of our method again. Enhancing SFIN's performance in scenarios involving extreme semantic bias will be our future work.

2) Results on DFC15-ML and MLRSNet: Due to the extremely high resolution, lower interclass similarity, and vast quantities, the difficulties of MLRSSC on DFC15-ML and MLRSNet have increased dramatically. Therefore, we study them together and present the numerical results of different models in Table II. It is easy to find that our SFIN achieves superior performance in all cases. For DFC15-ML, compared with the suboptimal model, the performance improvements

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TABLE II
PERFORMANCE OF DIFFERENT METHODS COUNTED ON DFC15-ML AND MLRSNET (%). THE BEST VALUES ARE IN BOLD, AND THE SECOND-BEST VALUES ARE UNDERLINED

Methods	DFC15-ML						MLRSNet					
	P_e	R_e	$F1_e$	P_c	R_c	$F1_c$	P_e	R_e	$F1_e$	P_c	R_c	$F1_c$
SNDL-BCE [50]	97.50	96.31	96.90	95.92	93.08	94.48	89.29	88.78	89.03	88.74	87.84	88.28
ResRF [11]	97.55	95.75	96.65	96.01	88.64	92.17	89.33	80.33	84.59	88.93	75.92	81.91
GCMLFNet [45]	97.50	96.86	97.18	95.38	94.62	95.00	87.53	86.42	86.98	87.78	87.08	87.43
AL [69]	96.46	97.28	96.87	94.93	94.33	94.63	87.37	88.76	88.06	87.29	86.71	87.00
ForestViT [61]	97.29	96.64	96.67	95.14	93.35	94.24	89.36	88.95	89.15	89.16	87.88	88.51
LD-GCN [27]	96.70	95.43	96.06	95.00	91.97	93.47	88.49	87.18	87.83	88.68	87.26	87.96
BoostLU [70]	97.45	96.92	97.17	95.36	93.16	94.25	89.16	89.22	89.19	88.65	87.94	88.29
SR-Net [26]	97.17	96.89	96.75	95.54	94.19	94.86	89.06	88.72	88.89	88.45	87.56	88.03
Ours	**97.57**	**97.36**	**97.47**	**96.04**	**94.76**	**95.40**	**89.37**	**89.35**	**89.36**	**89.25**	**88.01**	**88.63**

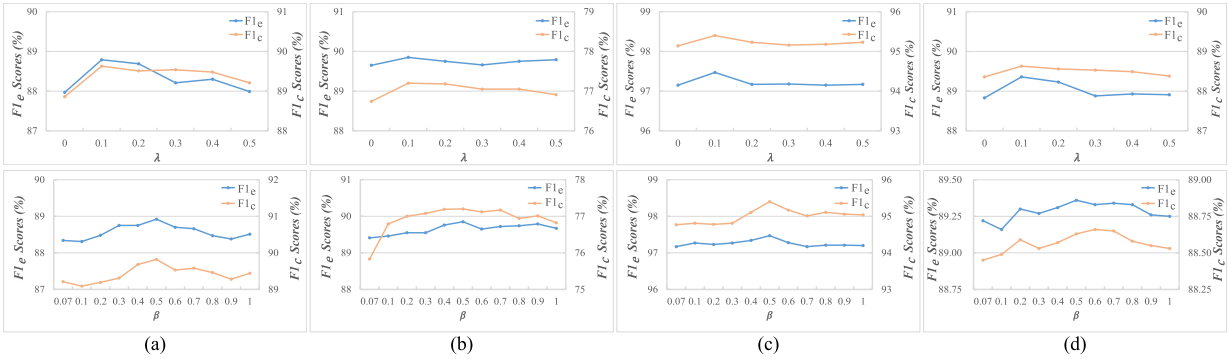


Fig. 7. $F1_c$ and $F1_c$ curves of the proposed SFIN with different λ (first line) and β (second line) counted on different datasets. (a) UCM-ML. (b) AID-ML. (c) DFC15-ML. (d) MLRSNet.

produced by SFIN are 0.02% (over ResRF in P_e), 0.08% (over AL in R_e), 0.29% (over GCMLFNet in $F1_e$), 0.03% (over ResRF in P_c), 0.14% (over GCMLFNet in R_c), and 0.40% (over GCMLFNet in $F1_c$). For MLRSNet, the performance enhancements are 0.01% (over ForestViT in P_e), 0.13% (over BoostLU in R_e), 0.17% (over BoostLU in $F1_e$), 0.09% (over ForestViT in P_c), 0.07% (over BoostLU in R_c), and 0.12% (over ForestViT in $F1_c$). These positive experimental results demonstrate that the proposed method is helpful in challenging datasets. A notable finding is that the performance of the compared models on the two datasets is unsteady. For instance, ForestViT performs better on MLRSNet than on DFC15-ML, and BoostLU is not good at handling DFC15-ML. In contrast, the behavior of our SFIN is stable in the two datasets, which proves its effectiveness and generalization once again.

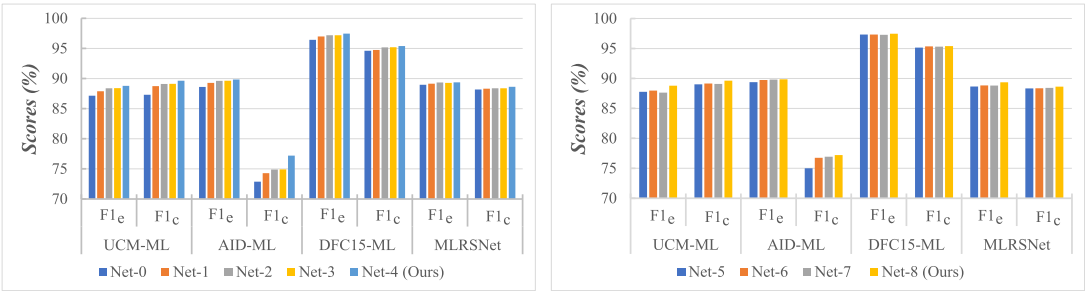
D. Parameter Analysis

In this section, we study the impact of λ and β on SFIN. Here, λ is used to control the contribution of the SB-CL term to overall loss, and β is the temperature coefficient for adjusting the model's sensitivity to sample similarity. To this end, we change their values one by one. The values of λ are varied from 0 to 0.5 with an interval of 0.1, and the values of β are tuned from 0.1 to 1.0 with an interval of 0.1. The case of $\beta = 0.07$ is also examined, which is a common value in contrastive learning. Again, we altered only one parameter in each experiment while keeping the other parameter fixed at the value mentioned in Section IV-B1. For clarity, only $F1_c$ and $F1_c$ are selected to count the results.

The curves of different SFINs with various λ counted on the four datasets are depicted in the top row of Fig. 7. By the observation, we can draw the following conclusions. First, across different datasets, the $F1_c$ and $F1_c$ scores are increased when λ changes from 0 to 0.1. This indicates that the SB-CL loss can enhance the model's discriminative ability for semantic information at both sample and category levels, thereby improving the final CLA results. Second, when λ continues to grow, the $F1_c$ and $F1_c$ scores of SFINs decline slowly. This is because the larger the λ, the more the model focuses on learning the semantic similarities between RS images and neglects CLA. Even so, the behavior of SFIN with $\lambda = 0.5$ is still stronger than that of SFIN with $\lambda = 0$, which confirms the effectiveness of the SB-CL loss. In sum, we suggest $\lambda = 0.1$.

The impacts of different β on SFIN counted on the four datasets are shown in the bottom row of Fig. 7. Observation results show that as β increases, SFINs' performance initially increases and then decreases, with the optimal value around 0.5. This shows that excessively increasing or reducing the model's sensitivity to the distance between semantics is not conducive to learning proper features for CLA. The reasons behind the above investigation are centered on the RS images, which exhibit high interclass similarity and intraclass diversity. A large β would overly smooth the model's similarity calculation, resulting in the embeddings of different semantics in the feature space being too close. Conversely, a small β would lead to the similar samples being far apart. Both of the above situations would lead to a decrease in CLA accuracy. In summary, we set the β value as 0.5.

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(a) (b)

Fig. 8. $F1_c$ and $F1_c$ bars of ablation studies counted on different datasets. (a) Ablation for DFEM, LSEM, and CIAM. (b) Ablation for the SS and SB-CL losses.

E. Ablation Studies

To analyze the contribution of each component to our SFIN, the ablation studies are conducted on four datasets in this section. Initially, we verify the effectiveness of each module. Then, we focus on analyzing the role of the SS and SB-CL losses. Similarly, only $F1_c$ and $F1_c$ are considered for clarity, and the experimental settings are consistent with the contents mentioned in Section IV-B1.

1) Effectiveness of Different Modules: As shown in Fig. 2, there are four modules in our SFIN: DFEM, LSEM, CIAM, and CM. DFEM is used to learn multiscale information from RS scenes. LSEM aims to extract semantic information, build their relationships, and supplement semantics into local convolutional feature maps. CIAM focuses on avoiding information loss through mutual exchange. CM produces the CLA results. To study their contributions, we construct the following networks.

1) *Net-0:* ResNet18 + classifier × 1.
2) *Net-1:* ResNet18 + LSET × 1 + classifier × 1.
3) *Net-2:* DFEM + LSET × 2 (LSEM) + classifier × 2 (CM).
4) *Net-3:* TFEM + LSET × 3 + classifier × 3.
5) *Net-4 (Ours):* DFEM + LSEM + CIAM + CM.

Net-0 is a standard ResNet18 model with a basic classifier. Net-1 incorporates an LSET to establish semantic associations using the output from the last residual layer of ResNet18. Net-2 represents SFIN without CIAM. Net-3 utilizes the last three residual layers of ResNet18 as a triple-scale feature extractor module (TFEM), followed by three parallel LSETs to enhance feature maps. The outputs of the three classifiers are then aggregated to produce the final CLA score. Notably, all the classifiers in the above networks share the same structure, consisting of a convolutional layer and a GAP layer, which is consistent with CM. Net-4 is actually our SFIN.

The performance of various networks counted on different datasets is shown in Fig. 8(a). We can draw the following conclusions. First, compared with Net-0, Net-1's $F1_c$ and $F1_c$ scores are higher, which illustrates the exploration of semantics and their associations are crucial for SFIN. This is mainly due to the following two points. One is that LSET's SSFD can accurately extract semantic feature vectors from feature maps. The other is AFUM in LSET can establish connections between different semantics and efficiently integrate them with feature pixels. Second, the behavior of Net-2 is stronger than that of Net-1, which proves that multiscale information contributes positively to MLRSSC. Third, excessive multiscale information actually limits the performance of SFIN. This phenomenon is particularly evident in the MLRSNet dataset, where the large volume amplifies the impact of redundant information, causing Net-3 to perform slightly worse than Net-2. In addition, no significant performance improvement is observed on other datasets. This aligns with the motivation behind our DFEM: while multiscale information is beneficial to MLRSSC, the influence of redundant information must be carefully managed. Therefore, in DFEM, we extract only the feature maps from the last two residual layers of ResNet18 and employ two LSETs in LSEM to extract semantic association information from these feature maps. Finally, Net-4's $F1_c$ and $F1_c$ scores are the highest, implying that CIAM can further enhance the feature representations by mutually interacting with the valuable knowledge across scales. To sum up, all modules in SFIN play a positive role.

To further explain how semantic relations affect SFIN, we visualize the feature maps extracted before and after LSEM in Fig. 9 in addition to the numerical results. It can be found that the salient areas in the feature maps obtained after LSEM are closer to the semantic labels. Taking the sample of AID-ML as an example (see second row), the salient regions in the feature map generated before LSEM concentrate on partial "Grass" and the surrounding areas. After LSEM, the enhanced feature map's salient regions still contain "Pavement," "Trees," and "Water." This change demonstrates that the semantic connections can improve SFIN's ability to extract various land covers.

2) Effectiveness of Different Losses: As mentioned in Section III-C1, our SFIN is jointly trained by the CLA, SS, and SB-CL losses. Because CLA is an indispensable loss function for MLRSSC, we first construct the following networks to study the contributions of the SS and SB-CL losses to SFIN.

1) *Net-5:* SFIN + CLA loss.
2) *Net-6:* SFIN + CLA loss + SS loss.
3) *Net-7:* SFIN + CLA loss + SB-CL loss.
4) *Net-8 (Ours):* SFIN + CLA loss + SS loss + SB-CL loss.

Net-5 means that only the CLA loss is used to train SFIN. Net-6 and Net-7 imply that we combine CLA with SS or

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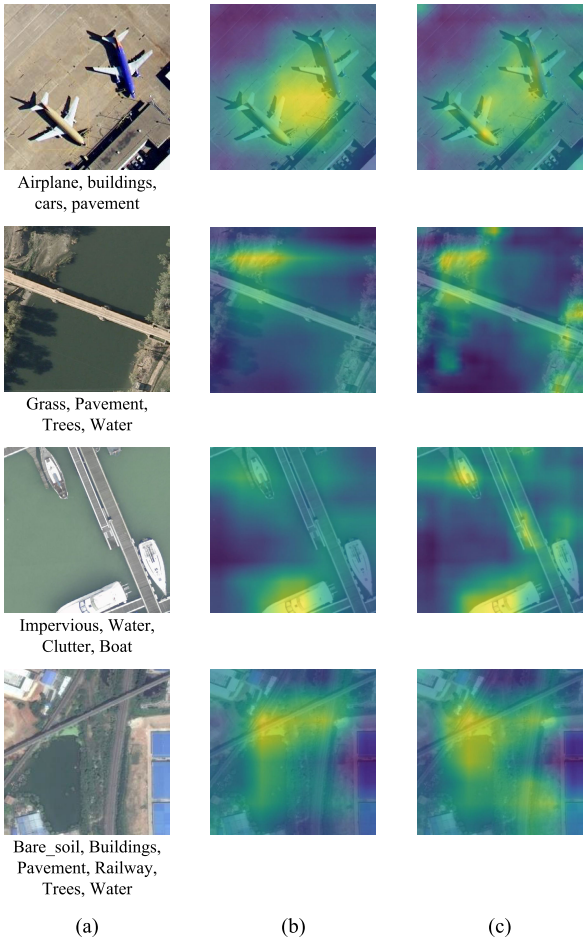


Fig. 9. Feature maps obtained before and after LSEM. The RS samples from the first to the fourth row are randomly selected from UCM-ML, AID-ML, DFC15-ML, and MLRSNet. (a) RS images and corresponding labels. (b) Feature maps obtained before LSEM. (c) Feature maps obtained after LSEM.

SB-CL to optimize the parameters of SFIN. Net-8 uses all loss functions to train SFIN.

The experimental results of different networks counted on the four datasets are summarized in Fig. 8(b). It is easy to find that as the loss functions are stacked, the model's performance gradually improves, which fully demonstrates that each loss function is playing its role. In detail, Net-5 generates the benchmarking results, which implies that the CLA loss's generalization. Net-6 and Net-7 outperform Net-5 indicating that maintaining the accuracy of semantics (SS loss) and enhancing their discrimination (SB-CL loss) can improve SFIN's ability to recognize complex land covers. When combining them together, Net-8 produces the best performance, which means the three losses provide interdependent clues for the final CLA. The above discussion confirms the usefulness of different losses.

In addition, we show the partial visualization results of class attention maps in Fig. 10. The first row displays the RS images, and the remaining rows are class activation maps of different semantic categories. We can find that the areas corresponding to different semantics can be highlighted exactly. This observation proves the accuracy of the semantics extracted by LSEM and further proves the effectiveness of SS loss. At the same

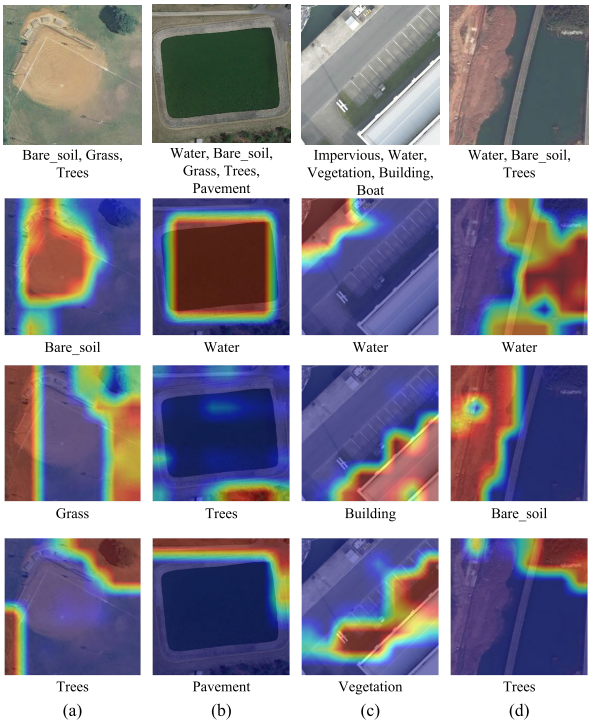


Fig. 10. Visualization results of class attention maps obtained from different datasets. The first row displays the RS images and the corresponding labels. Their class attention maps are shown in the remaining rows. (a) UCM-ML. (b) AID-ML. (c) DFC15-ML. (d) MLRSNet.

time, to further investigate the effectiveness of the SB-CL loss, we visualize the learned semantic features. For clarity, we only select the top seven semantics according to their quantity from each dataset and use the t-distributed stochastic neighbor embedding (t-SNE) algorithm [71] to map their semantic feature vectors (*DSF*) into 2-D space. Fig. 11 shows the feature distribution plot after dimensionality reduction. The figures in the first/second row are generated by SFIN with/without SB-CL loss. It is distinct that the clusters within the first row are more compact and separable than the clusters from the second row. This confirms our original intention of designing SB-CL loss: to bring closer/farther the distance between the same/different semantic feature vectors in the feature space.

F. Importance of Semantic Exploration

As mentioned in Section III, LSEM is developed to discover and analyze the semantics within RS images. It consisted of two identical LSETs, which are constructed based on ViT. Their functions are twofold: extracting the category-aware semantics and building their connections. To study the significance of LSEM and the explored semantics to SFIN, we replace LSETs in LSEM with other transformer encoders, including ViT [47], cross-head information interaction transformer (CIIT) [9], pyramid vision transformer-V2 (PVT-V2) [72], and efficient multiscale transformer (EMST) [4]. These SFIN variants are recorded as SFIN-ViT, SFIN-CIIT, SFIN-PVT-V2, and SFIN-EMST for convenience. Their $F1_c$ and $F1_c$ scores counted on the four datasets are summarized in Table III, and the results of SFIN are also displayed for reference. Note that all replaced

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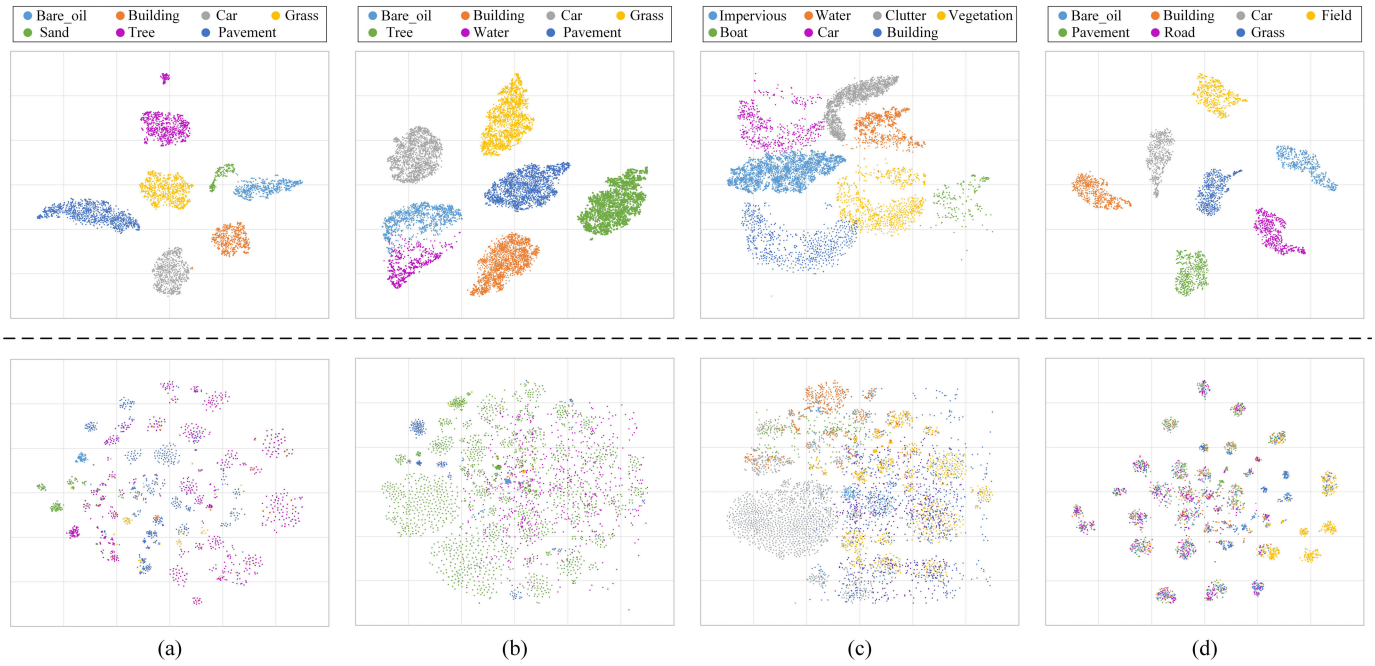


Fig. 11. 2-D feature distribution plot of semantic feature vectors generated by SFIN with SB-CL loss (first line) and without SB-CL loss (second line) using t-SNE algorithm on different datasets. (a) UCM-ML. (b) AID-ML. (c) DFC15-ML. (d) MLRSNet.

TABLE III

$F1_c$ (%), $F1_c$ (%), AND FLOPS (G) OF SFIN AND ITS VARIANTS COUNTED ON DIFFERENT DATASETS. THE BEST VALUES ARE IN BOLD, AND THE SECOND-BEST VALUES ARE UNDERLINED

Methods	UCM-ML		AID-ML		DFC15-ML		MLRSNet		FLOPs (G)
	$F1_e$	$F1_c$	$F1_e$	$F1_c$	$F1_e$	$F1_c$	$F1_e$	$F1_c$	
SFIN-ViT	88.30	89.53	89.75	74.51	97.28	95.17	88.51	88.56	2.787
SFIN-CIIT	88.14	89.31	<u>89.80</u>	75.11	97.34	<u>95.35</u>	88.44	<u>88.60</u>	2.788
SFIN-PVT-V2	<u>88.68</u>	89.05	89.16	76.08	<u>97.36</u>	95.33	88.86	88.45	2.770
SFIN-EMST	88.25	<u>89.56</u>	89.14	<u>76.16</u>	97.25	95.14	<u>89.32</u>	88.54	<u>2.751</u>
SFIN(Ours)	**88.79**	**89.63**	**89.85**	**77.20**	**97.47**	**95.40**	**89.36**	**88.63**	**2.711**

transformer encoders have four heads, the variants' experimental settings are the same as those discussed in Section IV-B1, and the SS and SB-CL losses are not involved in their training process.

We can get two points by observing Table III. First, the performance of all variants is good. This is because the long-distance dependencies that ViT, CIIT, PVT-V2, and EMST establish can help the model understand the complex contents within RS scenes. At the same time, the stability of DFEM, CIAM, and CM cannot be ignored. Second, SFIN achieves the best performance, which proves the importance of LSEM and the explored semantics for MLRSSC. Apart from $F1_c$ and $F1_c$ scores, we also count the floating point operations (FLOPs) of all networks, which are exhibited in the last column of Table III. It is easy to find that the calculation amount of SFIN is the smallest, further proving the effectiveness of our method.

G. Parameters and Time Costs

This section studies the parameters and inference time costs of SFIN. For clarity, we count the results using the UCM-ML dataset in Table IV. The results of the compared methods are also exhibited for reference. Through the observation, it is evident that SFIN features a relatively large number of parameters, primarily due to its adoption of a dual-branch structure for extracting multiscale information. In addition, the inference time costs of SFIN is not optimal. Fortunately, the gaps between SFIN and other superior methods (in terms of inference time costs and parameter volume) are acceptable. Lightening our model and speeding it up are our future works.

TABLE IV

INFERENCE TIME COSTS AND PARAMETER VOLUME OF DIFFERENT METHODS COUNTED ON UCM-ML

Methods	Inference Time Costs (ms)	Parameters (M)
SNDL-BCE [50]	7.14	11.24
ResRF [11]	7.16	11.18
GCMLFNet [45]	3.47	12.46
AL [69]	3.45	11.18
ForestViT [61]	4.69	29.11
LD-GCN [27]	3.84	13.82
BoostLU [70]	3.83	11.18
SR-Net [26]	3.86	15.34
Ours	4.01	16.25

V. CONCLUSION

In this article, we propose SFIN for MLRSSC. There are four modules in SFIN, i.e., DFEM, LSEM, CIAM, and CM. DFEM uses ResNet18 as the backbone and extracts multiscale features from RS images. LSEM first extracts global semantic information from feature maps and then establishes the relationship between local feature pixels and global semantics.

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This enables SFIN to understand RS image contents from the perspective of semantic connections. CIAM interacts with information at different scales to avoid information loss caused by scale differences. At the end of SFIN, CM produces the CLA results. In addition, we design SS and SB-CL losses to train SFIN with the common CLA loss jointly. SS loss is used to extract the correct semantic information, thereby ensuring the establishment of correct semantic connections. SB-CL loss is a semantic-level contrastive learning loss function. By narrowing/distancing the distances between the same/different semantics in the feature space, the SFIN's feature extraction capabilities for different land covers are enhanced. The encouraging results of the four datasets illustrate the usefulness of SFIN.

The uniqueness of our SFIN lies in: 1) combining feature learning with semantic relationship building to help the model fully understand complex contents within RS scene images and 2) ensuring the accuracy and discriminability of semantic information through deep supervision paradigms and the improved semantic-level contrastive learning loss function.

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